

Pro-inflammatory NF- κ B and early growth response gene 1 regulate epithelial barrier disruption by food additive carrageenan in human intestinal epi...

The widely used food additive carrageenan (CGN) has been shown to induce intestinal inflammation, ulcerative colitis-like symptoms, or neoplasm in the gut epithelia in animal models, which are also clinical features of human inflammatory bowel disease. In this study, the effects of CGN on pro-inflammatory transcription factors NF- κ B and early growth response gene 1 product (EGR-1) were evaluated in terms of human intestinal epithelial barrier integrity. Both pro-inflammatory transcription factors were elevated by CGN and only NF- κ B activation was shown to be involved in the induction of pro-inflammatory cytokine interleukin-8. Moreover, the integrity of the in vitro epithelial monolayer under the CGN insult was maintained by both activated pro-inflammatory transcription factors NF- κ B and EGR-1. Suppression of NF- κ B or EGR-1 aggravated barrier disruption by CGN, which was associated with the reduced gene expression of tight junction component zonula occludens 1 and its irregular localization in the epithelial monolayer. Copyright © 2012 Elsevier Ireland Ltd. All rights reserved.